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W8-01: BigTable

COMPSCI4064 (H) and COMPSCI5088 (M)

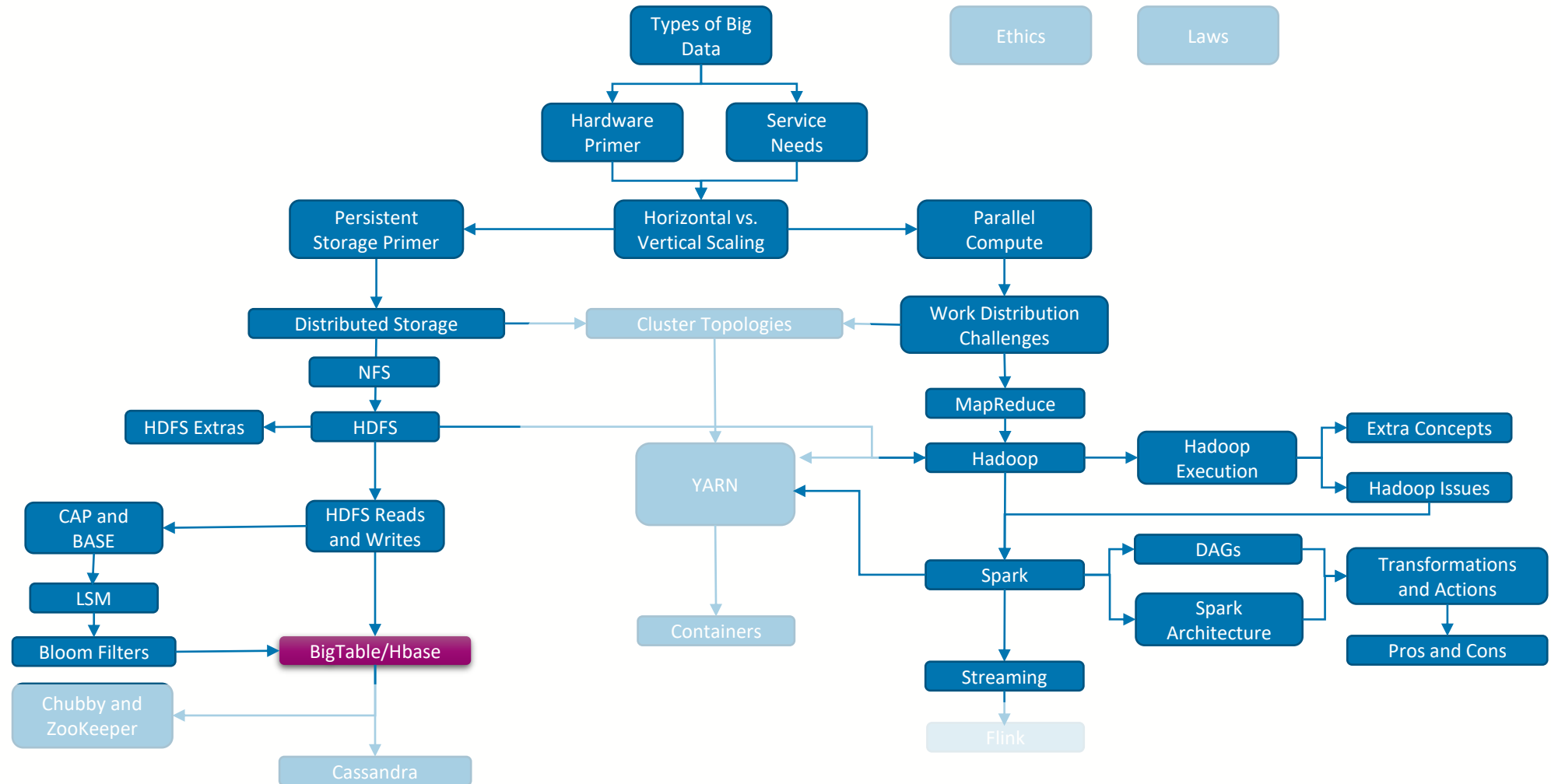
Dr. Richard McCreddie

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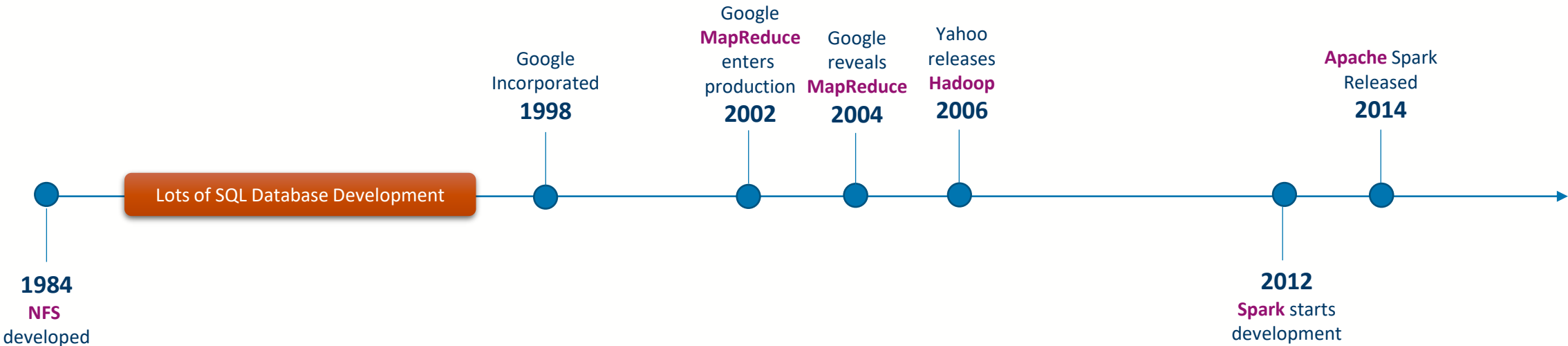


Where are we?



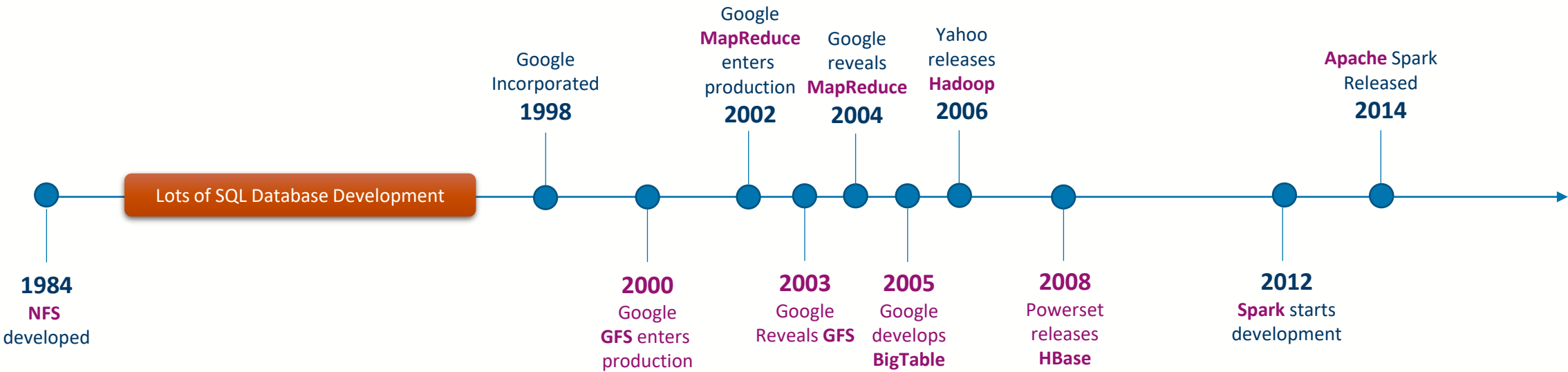


Technology Timeline





Technology Timeline





Google... Innovating Behind the Scenes Again

- Between 2000 and 2004, the Google Index grew over 20x
- In 2004 Google submitted a patent for a new invention and also published an associated research paper:

MapReduce: simplified data processing on large clusters

[J Dean](#), [S Ghemawat](#) - Communications of the ACM, 2008 - dl.acm.org

... the performance of **MapReduce** on two computations running on a **large cluster** of machines. One computation searches through approximately one terabyte of **data** looking for a particu...

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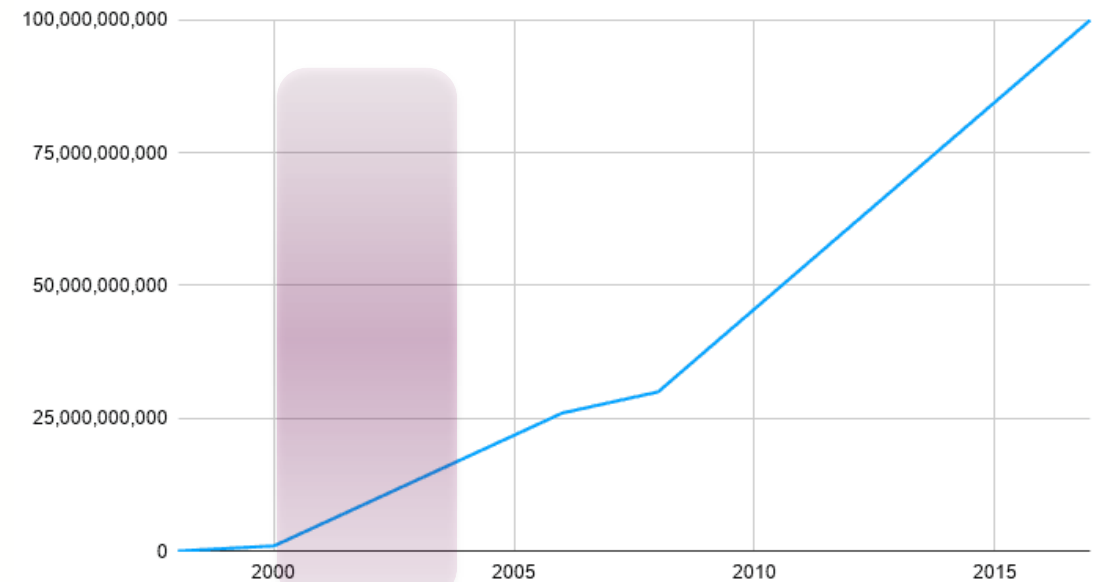
[HTML] MapReduce: Simplified data processing on large clusters

[J Dean](#), [S Ghemawat](#) - 2004 - usenix.org

... Our implementation of **MapReduce** runs on a **large cluster** of ... **MapReduce** computation processes many terabytes of **data** on ... the system **easy** to use: hundreds of **MapReduce** programs ...

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What did Google work on next?



Google... Innovating Behind the Scenes Again

- After inventing MapReduce, their solution to horizontal scaling of compute, their engineers started work on a **better solution for managing distributed data**
 - At this time Google already has an HDFS equivalent (well before HDFS was invented), referred to as the **Google File System (GFS)**
 - Like the HDFS, GFS is **fast for large block reads**, but **slow for file writes** and append operations
 - They wanted a faster solution for **writing data**
- Solution: **BigTable**



Google BigTable

- As its name suggests BigTable is a **Table storage solution**
 - Rather than a file storage solution like NFS, GFS or HDFS
 - This shift from thinking about data storage in terms of files to thinking in terms of Tables is important, and is still with us today – think Spark RDDs or Python DataFrames
- Four core high-level ideas here:
 - Most big data processes want to **read in and later write out a contiguous set of records**, similar to rows in a table
 - **Records** are usually **structured into standard fields**, which could be columns in a database table (although not all records will have all fields)
 - Unlike SQL databases, we normally **don't have explicit relationships between fields** - no need for a draconian relational database management system (RDMS)
 - We may want to **store multiple versions of a record** representing it at different points in time.

Google BigTable

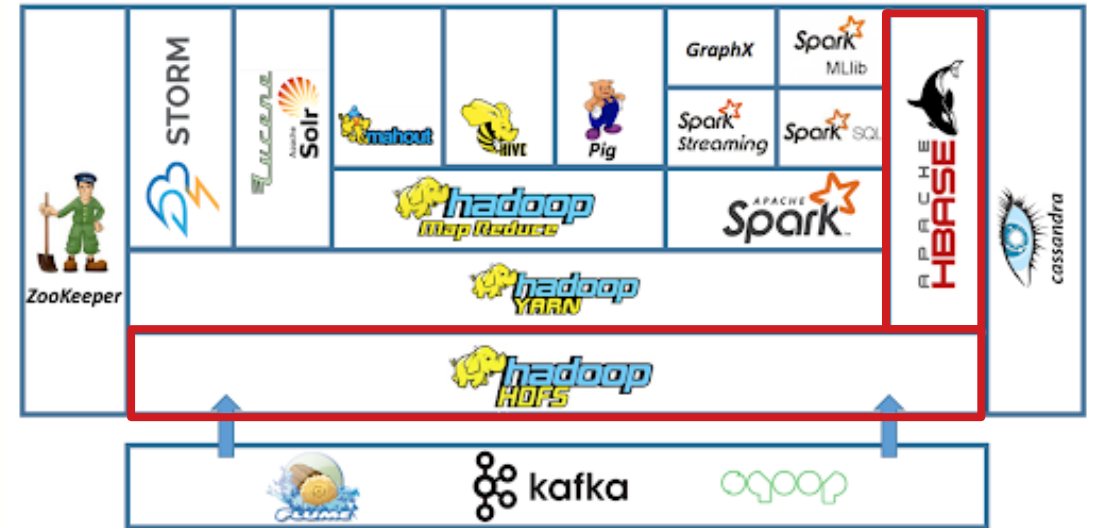
- **Sparse**: Records (rows) can be incomplete (only contain a subset of the fields)
- **Distributed**: Sets of records are stored as blocks and distributed amongst multiple machines
- **Persistent**: Records are always stored to persistent storage
- **Multidimensional**: Data is indexed through a combination of the row key, column name, and timestamp
- **Sorted**: Records are ordered lexicographically by the RowKey
- **Map**: It's a map between RowKeys and Records

RowKey	Name	Degree	Timestamp
000001	David	MSc CS	21/11/2022...
000002	Jing	MSc CS	21/11/2022...
000003	Xi	MSc DS	21/11/2022...
000004	John	MSc DS	29/11/2022...
000005	Ellen		21/11/2022...
000006	Yang		21/11/2022...
000007	Misbah	MSci Joint	21/11/2022...
000008	Simona	MSci	26/11/2022...



HBase

- Google, as usual did not release an open-source version of BigTable, so it was up to someone else to produce an alternative, this time it was a company called PowerSet
 - They released **HBase** in 2008
 - Its basically BigTable running on top the HDFS for underlying file storage





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